Run report: run-20261011T2330Z-gls-validate

What ran. `./target/release/examples/gls_loop_validate --pbits 127 --points 24 --seed 20261012 --discs -619,-23,-71,-431,-3019 --json research/isogeny_loop_glv_20261012/gls_validate.json` (in `isogeny_algos/`). Source: working tree that became checkpoint commit 54a16f6d4 on branch `claude/isogeny-loop-glv-sweeper`. rustc 1.97.0, release profile, Linux 6.18 x86_64 cloud container, not isolated. Start 2026-10-11T23:30Z (`../../validate_run_start_utc.txt`); wall time 2.8 s (`../../gls_validate.log`).

Replay. On 2026-10-12 the same command was re-run on the final source, after helper functions moved into `src/gls_loop.rs` and the clippy-only edits. The Markdown table was byte-identical, and the JSON differed only in its recorded `command` (output path).

Arms. One arm per discriminant, D in {−619, −23, −71, −431, −3019}. Each runs these steps:
- certified cheapest loop;
- instance search ($4p = t^2 + |D|v^2$, $r = \#E'(F_{p^2})/h$ prime, $p \equiv 3 \bmod 4$);
- H_D by the analytic method, roots mod p, curve, and a twist if needed;
- crater walks;
- an F_p check on $E(F_p)[r_E]$;
- the ψ and twisted-loop checks on 24 points of order r on E′.

Runs requested / completed. 5 / 5, exit 0, verdict ALL PASS. No failures.

Recorded values (`../../gls_validate.md`, `../../gls_validate.json`):

D	steps	H_D deg / bits / worst err	$\log_2 r$	ψ	loops act as	verdict
−619	5,5,7	5 / 303 / 2^{-90}	251	+μ 24/24	−λ(s); −λ(−s)	PASS
−23	2,3	3 / 150 / 2^{-86}	251	+μ 24/24	−λ(−s); −λ(s)	PASS
−71	2,3,3	7 / 238 / 2^{-92}	248	+μ 24/24	−λ(−s); −λ(s)	PASS
−431	2,2,3,3,3	21 / 631 / 2^{-118}	252	+μ 24/24	+λ(−s); +λ(s)	PASS
−3019	5,5,5,13	7 / 523 / 2^{-88}	253	+μ 24/24	−λ(−s); +λ(s)	PASS

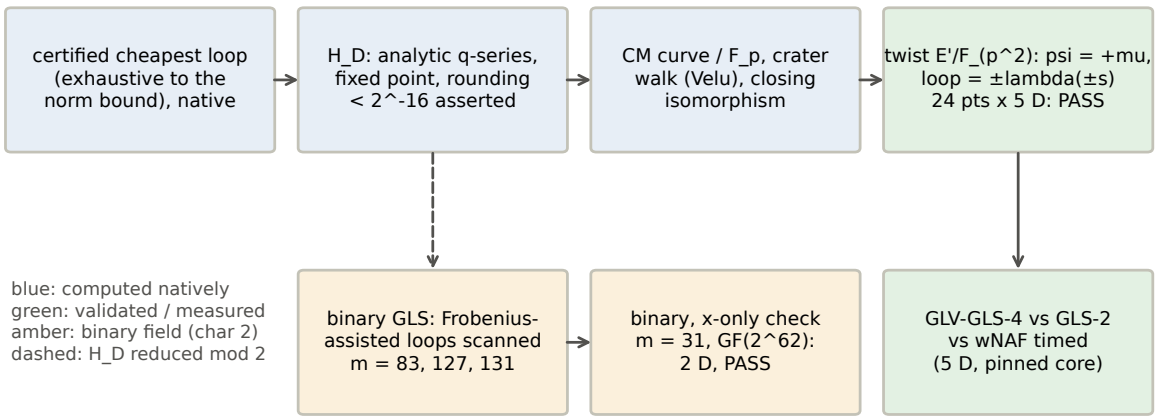


Figure 1: What this run validates (green, top right) within the round-3 pipeline. Diagram, not data: each row has a single pass/fail value, so there are no comparable quantities to plot.

Observations only. Interpretation: `../../README.md` §2.